MSG No. 52.317 P_Anna [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[2,3,9,12]`
3	$\left[\frac{3}{4}, 0, 0\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,7,13,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[2,3,9,12]`
3	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[6,7,13,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2'/m'..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,10,14]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,6,9,13]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2}, 0, 0\right]$	`[4,8,11,15]`

Table 4: Wyckoff site: `4d`, site symmetry: `2'/m'..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,5,10,14]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[2,6,9,13]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,7,12,16]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8,11,15]`

Table 5: Wyckoff site: `4e`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[3,4,15,16]`
3	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,8,11,12]`

Table 6: Wyckoff site: `4f`, site symmetry: `2/m’..`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,4,15,16]`
3	$\left[0,\ \frac{3}{4},\ \frac{3}{4}\right]$	`[5,6,9,10]`
4	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[7,8,11,12]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ z\right]$	`[1,4]`
2	$\left[\frac{1}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[2,3]`
3	$\left[\frac{3}{4},\ 0,\ -z\right]$	`[5,8]`
4	$\left[\frac{3}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{4},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,12]`
6	$\left[\frac{1}{4},\ 0,\ -z\right]$	`[10,11]`
7	$\left[\frac{3}{4},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,16]`
8	$\left[\frac{3}{4},\ 0,\ z\right]$	`[14,15]`

Table 8: Wyckoff site: `8h`, site symmetry: `.2’.`

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ 0\right]$	`[1,11]`
2	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[2,12]`
3	$\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[3,9]`
4	$\left[\frac{1}{4},\ -y,\ 0\right]$	`[4,10]`
5	$\left[\frac{3}{4},\ -y,\ 0\right]$	`[5,15]`
6	$\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	`[6,16]`
7	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	`[7,13]`
8	$\left[\frac{3}{4},\ y,\ 0\right]$	`[8,14]`

Table 9: Wyckoff site: `8i`, site symmetry: `2’..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,10]`
2	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,9]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,12]`
4	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	`[4,11]`
5	$[-x,\ 0,\ 0]$	`[5,14]`
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[6,13]`
7	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,16]`
8	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	`[8,15]`

Table 10: Wyckoff site: 8j, site symmetry: 2’..

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,10]
2	$\left[x,\ 0,\ \frac{1}{2}\right]$	[2,9]
3	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	[3,12]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	[4,11]
5	$\left[-x,\ \frac{1}{2},\ 0\right]$	[5,14]
6	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[6,13]
7	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	[7,16]
8	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[8,15]

Table 11: Wyckoff site: 8k, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[1,2]
2	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[3,4]
3	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[5,6]
4	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{3}{4}\right]$	[7,8]
5	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[9,10]
6	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{3}{4}\right]$	[11,12]
7	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[13,14]
8	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{4}\right]$	[15,16]

Table 12: Wyckoff site: 8l, site symmetry: m’..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,14]
2	$\left[0,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[2,13]
3	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[3,16]
4	$\left[\frac{1}{2},\ -y,\ z\right]$	[4,15]
5	$[0,\ -y,\ -z]$	[5,10]
6	$\left[0,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[6,9]
7	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[7,12]
8	$\left[\frac{1}{2},\ y,\ -z\right]$	[8,11]

Table 13: Wyckoff site: 16m, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[2]
3	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[6]
7	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[7]

continued ...

Table 13

No.	position	mapping
8	$\left[x+\frac{1}{2},\, y,\, -z\right]$	[8]
9	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$[x,\, -y,\, -z]$	[10]
11	$\left[\frac{1}{2}-x,\, y,\, -z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[12]
13	$\left[-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[13]
14	$[-x,\, y,\, z]$	[14]
15	$\left[x+\frac{1}{2},\, -y,\, z\right]$	[15]
16	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[16]